

0.1 Radial Schrödinger

We first verify the accuracy of the spectral finite-element framework for the all-electron radial Schrödinger equation, which is obtained by setting $\hat{V}_{xc} = \tilde{V}_H = 0$ in the radial Kohn–Sham equation and admits analytical solutions. With a pure Coulomb nucleus $V_{\text{nuc}}(r) = -Z/r$ and Hartree and exchange–correlation terms omitted, the radial eigenvalue problem is

$$\left(-\frac{1}{2} \frac{d^2}{dr^2} - \frac{Z}{r} + \frac{l(l+1)}{2r^2}\right) \tilde{R}_{nl} = \lambda_{nl} \tilde{R}_{nl}, \quad (1)$$

with hydrogenic levels $\varepsilon_n = -Z^2/(2n^2)$ Ha. We consider $Z = 92$ and compute all occupied eigenvalues using a domain size of $R_{\text{max}} = 40$ Bohr, an exponential mesh with shift $c = 0$ and concentration parameter $a = 100$, $N_{fe} = 12$ finite elements, polynomial degree $p = 20$, and quadrature order $q = 60$. Figure 1 shows the eigenvalue errors relative to the analytical values, demonstrating accuracies better than 10^{-10} Ha and confirming the accuracy of the spectral finite-element framework. The finite-element eigenvalues λ_{nl} and their comparison to ε_n are listed in Table 1.

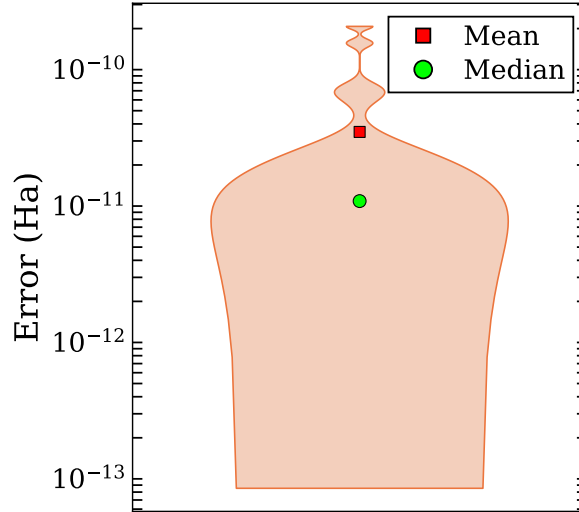


Figure 1: Distribution of eigenvalue errors for the radial Schrödinger equation corresponding to $Z = 92$.

Table 1: Occupied subshell eigenvalues for $Z = 92$ (Ha); analytic $\varepsilon_n = -Z^2/(2n^2)$.

n	l	g_{nl}	numeric (Ha)	analytic (Ha)	Error (Ha)
1	0	2	-4231.9999999998436	-4232.0000000000000	1.56×10^{-10}
2	0	2	-1057.9999999997924	-1058.0000000000000	2.08×10^{-10}
2	1	6	-1058.0000000000209	-1058.0000000000000	2.09×10^{-11}
3	0	2	-470.2222222222873	-470.2222222222222	6.50×10^{-11}
3	1	6	-470.2222222221507	-470.2222222222222	7.15×10^{-11}
3	2	10	-470.2222222222160	-470.2222222222222	6.20×10^{-12}
4	0	2	-264.5000000000234	-264.5000000000000	2.34×10^{-11}
4	1	6	-264.5000000000005	-264.5000000000000	5.12×10^{-13}
4	2	10	-264.5000000000092	-264.5000000000000	9.15×10^{-12}
4	3	14	-264.4999999999808	-264.5000000000000	1.92×10^{-11}
5	0	2	-169.2800000000126	-169.2800000000000	1.26×10^{-11}
5	1	6	-169.2799999999855	-169.2800000000000	1.45×10^{-11}
5	2	10	-169.2800000000001	-169.2800000000000	8.53×10^{-14}
5	3	14	-169.2799999999980	-169.2800000000000	2.05×10^{-12}
6	0	2	-117.5555555555597	-117.5555555555556	4.11×10^{-12}
6	1	6	-117.5555555555492	-117.5555555555556	6.35×10^{-12}
6	2	10	-117.5555555555598	-117.5555555555556	4.21×10^{-12}
7	0	2	-86.3673469387809	-86.3673469387755	5.37×10^{-12}